

Quick, repeatable, real-time data.

According to Erkan Ickam of EMSCAN, a leading developer of fast magnetic very-near-field measurement applications, "Traditional EMC far-field techniques measure the emissions levels; they are not useful for small, printed circuit board assembly (PCBA) level diagnosis as they can't point at the source of emissions on the PCB." The very-near-field technique is fast and repeatable; interaction with DUT is unavoidable in this area. "It can measure the entire PCB continuously to make sure intermittent events are captured even when they occur in areas of the board that are not expected to radiate," he adds.

In other words, very-near-field technique is a good method to quickly identify localised hot spots at a PCBA level. It lets you identify frequency content of emissions and make spatial maps of the currents at each frequency as they occur on DUT. These spatial maps give information to test engineers about where the ultimate source of EMC emissions are, at the PCB level. These very-near-field spatial maps cannot be directly compared to any existing EMC standards like IEC and ANSI; they, however, provide insight into troubleshooting or root-cause analysis for any standard.

Complementary to all standards. Very-near-field testing is a pre-compliance tool and there are no standards for EMC pre-compliance. An example is the EMxpert by EMSCAN. Regardless of the standard, when current EMC chamber measurements indicate a failure, the test results do not generally indicate what the cause of the problem is. This is where a quick measurement utilising very-near-field is extremely valuable. It will indicate the source of emissions, coupling paths, resonant points, shielding issues as well as many other potential problems. This can help the engineers track down the problem quicker and make proper changes to the design.

Indoor testing. Another advantage cited for near-field measurement techniques is that testing can be accomplished indoors, eliminating problems due to weather, electromagnetic inter-

Latest EMC test equipments

2171B Bore sight antenna tower for ANSI standard C63.4 by ETS-Lindgren. Released in the first quarter of 2014, model 2171B antenna tower is designed to meet ANSI standard C63.4 requirements for measurements above 1GHz. The patented Bore sight system keeps the antenna aimed at the DUT during the antenna's ascent or descent along the antenna mast.

Red Edge series for electromagnetic pulse protection by ETS-Lindgren. This product line, announced in the last quarter of 2013, helps negate high-altitude electromagnetic pulse (HEMP) and intentional electromagnetic interference (IEMI) threats.

Double-ridged waveguide horn and BiConiLog antennae 3116C-PA and 3117-PA with matched preamplifiers by ETS-Lindgren. The antennae are available with matched preamplifier since the third quarter of 2013, both calibrated as a single unit. All the possible mismatches between the antenna and the preamplifier would be taken into account in the antenna factor (AF) which is provided with each unit.

CEMS100 test platform for EMS (electromagnetic susceptibility) testing for IEC/EN 61000-4-3 by Rohde & Schwarz. This is a flexible and reliable off-the-shelf solution for EMS measurements in line with IEC/EN 61000-4-3 standard, announced during the first quarter of 2014. The platform includes all the components needed for EMS and EMI measurements, covers common frequency ranges and field strengths required for pre-compliance tests and certification.

ESRP EMI test receiver by Rohde & Schwarz. The device has EMI test receiver and signal/spectrum analyser combined in one box, and offers optional pre-selection and preamplifier. It was released during the first quarter of 2014.

Line impedance stabilisation networks (LISNs)/artificial mains networks (AMNs) series by TDK RF. This new line of LISNs/AMNs, announced in the first quarter of 2014, can be used for various conducted EMI testing applications. These are offered separately as well as integrated into compliant and pre-compliance solutions for automated EMI measurements.

MODS3000 series of mixed-domain oscilloscopes (with integrated spectrum analyser) by Tektronix. This customisable and upgradable 6-in-1 integrated oscilloscope includes integrated spectrum analyser, logic analyser, protocol analyser, arbitrary function generator and digital voltmeter/counter. It was released in the first quarter of 2014.

RSAS5000 spectrum analyser by Tektronix. A mid-range spectrum analyser released during the last quarter of 2013. It has features like DPX Live RF spectrum display, seamless data capture, and automatic pulse measurement and detection.

ference, security concerns, etc.

Drawbacks. Each hot-spot seen in very-near-field probing is an independent vector by itself. Unless the magnitude and polarity of these independent vectors are quantified, convergence to real-lab measurements cannot be assured. Another issue is that the DUT must be smaller than the scanner for pre-compliance. The device should be disassembled for testing.

Testing at the component level

P. Chow Reddy suggests that, to test at component level, make a simple loop probe with a coaxial cable that could be connected to the spectrum analyser. One end of the cable can be stripped and centre wire extended to the braid of the shielded wire. This forms a complete loop with respect to the shield. This arrangement picks up radiations in the loop and transfers to

MAJOR CONTRIBUTORS TO THIS STORY

- Erkan T. Ickam, director of marketing, EMSCAN
- Kenneth Wyatt, senior EMC engineer, Wyatt Technical Services, LLC
- P. Chow Reddy, manager-R&D (Power Division), ICOMM Tele
- Raghu Rao, manager of application engineering team, India, Tektronix
- Rajneesh Raveendran, project manager, Tarang Lab, Wipro
- Vishal Gupta, application expert, Agilent Technologies India

spectrum analyser and hence gives a rough picture of the radiation level.

To specifically identify the radiating component, strip a thin-shielded wire with braid, make the centre wire into a small loop and solder it on to the braid. Then drop silicon sealant on the loop. This forms a mini probe. With one end connected to a spectrum analyser, this probe can be brought close to the specific components to analyse the pickup radiations.

Antennae tuned to particular range of frequencies and resonances are readily available in the market. These can be used to get a very brief picture of the radiations in open labs. ●

The author is a technical correspondent at EYF